MPG No. 12.4.43 $42'2'$ [Type III, tetragonal] [T tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & T_z \end{bmatrix}$$

$$T_z = \mathbb{T}_z^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ -T_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{M}_z^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & T_{xxz} \\ 0 & 0 & T_{zzz} \\ 0 & T_{xxz} & 0 \\ T_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)}$$
$$T_{zzz} = 3\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & T_{xxz} \\ 0 & 0 & 0 \\ 0 & -\frac{T_{xxz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxz} = 2\mathbb{T}_z^{(2)}$$
$$\frac{T_{xxz}}{2} = \mathbb{T}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & T_{yzy} & 0 \\ -T_{yzy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzy} = \mathbb{T}_z^{(3)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & -T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ T_{xxxy} & -T_{xxxy} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = \mathbb{T}_{4z}^{\alpha(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & -T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & -T_{yzzx} & 0 & 0 \\ -T_{xxxy} & T_{xxxy} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = 2\mathbb{M}_z^{(1)} - 2\mathbb{M}_z^{\alpha(1)}$$
$$T_{yzzx} = -\mathbb{M}_z^{(1)} - 4\mathbb{M}_z^{\alpha(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & T_{yzzx} & 0 \\ -T_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = -\mathbb{M}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxxy} \\ 0 & 0 & T_{xxxy} \\ 0 & 0 & T_{zzxy} \\ 0 & T_{yzzx} & 0 \\ T_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = \mathbb{M}_z^{(3)} - \mathbb{M}_z^{\alpha(2)}$$
$$T_{zzxy} = \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{(4)} + 2\mathbb{M}_z^{\alpha(2)}$$
$$T_{yzzx} = \mathbb{M}_z^{(4)} - \mathbb{M}_z^{\alpha(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & T_{yzzx} & 0 & 0 \\ T_{xyxx} & T_{xyxx} & T_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)}$$
$$T_{xyxx} = \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)}$$
$$T_{xyzz} = \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxy} & 0 & 0 & T_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & -T_{xxyx} & 0 & 0 & -T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & T_{zzxy} & 0 & 0 & -T_{zzxy} \\ 0 & 0 & 0 & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & -T_{yzxz} & 0 & 0 & -T_{yzzx} & 0 & 0 \\ T_{xyxx} & -T_{xyxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = 2\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + \mathbb{T}_{4z}^{\alpha(1)}$$
$$T_{xxyx} = 2\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + \mathbb{T}_{4z}^{\alpha(1)}$$
$$T_{zzxy} = \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{(4)} + 2\mathbb{M}_z^{\alpha(2)}$$
$$T_{yzzx} = -\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)}$$
$$T_{yzxz} = -\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)}$$
$$T_{xyxx} = -2\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)} + \mathbb{T}_{4z}^{\alpha(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{yzxz} & 0 & 0 & T_{yzzx} & 0 & 0 \\ T_{xyxx} & T_{xyxx} & T_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)}$$
$$T_{yzxz} = \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)}$$
$$T_{xyxx} = \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)}$$
$$T_{xyzz} = \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & T_{xxxy} & 0 \\ -T_{xxxy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{xxxy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -T_{xxxy} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = \mathbb{T}_{4z}^{\alpha(1)}$$